

ColorAptitude — CAP-W10

A digital platform for professional colour competency assessment based on a three-layer cognitive framework, OKLCH-based stimulus generation, and differentiated per-layer validation

Technical Whitepaper – Scientific Disclosure & IP Positioning Status: Publication-ready / Prior-art disclosure **Document code:** CAP-W10 v1.0 **Author:** Markus Kotterink **Managing entity:** Xconea — BNK Foundation, Netherlands

1. Abstract

This document describes a computer-implemented digital platform for professional colour competency assessment, training, and certification. The platform is designed for the qualification of visual observers engaged in colour-critical professional work — automotive coatings, textile production, graphic arts, food and cosmetic quality control, conservation, and related domains — and is positioned as an occupational-qualification instrument in the sense of ASTM E1499-16 rather than as a clinical screening or grading test in the sense of the FM100, Cambridge Colour Test, or Colour Assessment and Diagnosis test.

The platform is organised around a **three-layer cognitive framework** in which professional colour competency is decomposed into three empirically distinguishable capabilities: **discrimination** (detection that a colour difference exists), **attribution** (identification of which perceptual dimension has changed), and **communication** (translation of the percept into standardised professional terminology). The three layers rest on independent empirical foundations in the perceptual, cognitive, and linguistic-professional literature. They are measured by dedicated instruments, scored through independent pipelines, and reported as separate scalar or vector values that are preserved through all downstream processing.

A **three-by-three competency matrix** combines the three layers with the three perceptual colour dimensions (hue, lightness, chroma), producing nine resolved competency cells per assessment session. The matrix preserves cross-axis patterns that single-score or single-axis reporting would collapse, and supports targeted training-module selection, cell-resolved certification scope, and cross-session progress tracking at cell granularity.

Stimulus generation uses **OKLCH with per-hue gamut-aware trial construction**, ensuring that each trial varies exactly one perceptual dimension without cross-dimensional leakage due to gamut clipping. A **two-path display calibration architecture** accommodates both observer-managed deployment (software-only Path A) and ASTM E1499-16 §5.3 compliant qualification (hardware-measured Path B), with unified composite scoring across paths and certificate-tier annotation preserving the calibration basis in downstream verification.

Validation of the three competency layers follows a **differentiated three-track strategy** that reflects their distinct psychometric development stages: criterion-referenced normative data collection for discrimination, known-groups validity through expert-panel contrast for attribution, and longitudinal construct-validity monitoring paced by curriculum deployment for communication. This differentiated strategy is disclosed as an integral component of the architecture, not as a posthoc validation protocol.

The subject matter disclosed in Sections 3.1 through 3.8 includes the three-layer framework, the 3×3 matrix reporting architecture, the differentiated validation strategy, a five-segment 18-cap Hue Ordering Diagnostic with three red-green opponent segments, the two-path calibration architecture, cardinal-nearest hue classification avoiding Logvinenko circularity, OKLCH gamut-aware trial generation, and Kuehni-anchored closed-set attribution scoring. A companion scientific manuscript describing the same subject matter in its academic context has been prepared separately for submission to *Colour Research and Application*.

2. TECHNICAL BACKGROUND

This section establishes the minimum context required to read Sections 3.1 through 3.8 without reference to external literature. It is deliberately compact. Readers seeking fuller treatment are referred to the companion scientific manuscript.

2.1 Professional colour competency as a three-part capability

Professional colour judgment in industrial contexts — automotive coatings, textiles, graphic design, food and cosmetic quality control, and related domains — requires three capabilities that are empirically distinguishable:

Discrimination. The ability to detect that a colour difference exists. This is a perceptual capability, grounded in retinal and early cortical processing, and well documented in the clinical colour-vision literature.

Attribution. The ability to identify which perceptual dimension (hue, lightness, or chroma) has changed, and in which direction. This is a cognitive-analytical capability. Neuropsychological evidence (double dissociation following focal brain lesions) and developmental evidence (pre-verbal infant categorical perception) establish this layer as separable from discrimination at the neural level.

Communication. The ability to translate perceived colour differences into standardised professional terminology, and to reconstruct a percept from a description. This is a linguistic-professional capability. Cross-cultural colour-naming research and colour-naming stability across the ageing lifespan establish this layer as separable from both discrimination and attribution.

Existing clinical colour-vision instruments (FM100, Cambridge Colour Test, Colour Assessment and Diagnosis test) measure discrimination only. No commercially available instrument known to the author combines measurement of all three capabilities in an integrated assessment platform before the publication of this document.

2.2 Colour dimensions and perceptually uniform colour spaces

Three colour dimensions are inherent in any colour description in a perceptually uniform colour space: **hue** (angular position on the colour circle), **lightness** (position on the black-to-white axis), and **chroma** (departure from neutral grey at constant lightness). Each dimension can vary independently under appropriate stimulus control.

The HSB (Hue–Saturation–Brightness) and legacy sRGB colour spaces are not perceptually uniform: equal numerical steps do not correspond to equal perceived differences. CIELAB is approximately uniform but exhibits hue non-uniformities in the blue region. The OKLCH colour space (Ottoosson, 2020), derived through a CAM16-aware non-linear transformation of Oklab, provides perceptual uniformity sufficient for single-dimension trial generation across hue, lightness, and chroma. OKLCH is used throughout the methods disclosed in Section 3; OKLCH itself is prior art of Ottoosson and is not claimed in this publication.

2.3 ASTM E1499-16 as a qualification framework

ASTM E1499-16 *Standard Guide for the Selection, Evaluation, and Training of Observers* specifies observer-qualification requirements across selection, baseline evaluation, training, formal qualification, retraining, and re-qualification. Sections relevant to the methods disclosed in this document include §5.3 (compliance requirements for test instruments), §6.3 (colour discrimination testing), §6.4 (magnitude scaling), §7.1.1 (warm-up observations), §7.2 (training integration), and §8.1.1 (session limits). The standard is prior art and is not claimed. Implementation of the standard's requirements in a specific integrated digital platform, in combination with the novel methods disclosed in Section 3, is the subject of the present publication.

2.4 Conventions used in Section 3

Each of the eight innovations disclosed in Section 3 follows the same internal structure:

Problem addressed. The technical or methodological problem the innovation solves.

Prior art and distinction. The known prior art, and the specific distinction between the prior art and the claimed subject matter.

Claimed subject matter. The disclosure of the innovation, with a distinction between broad claim (the general principle) and narrow claim (the specific implementation) where relevant.

Enabling description. Technical detail sufficient for a person skilled in the art of digital colour assessment to implement the innovation without further invention.

Embodiments. Specific implementation variants, including parameter values, algorithms, and data structures where applicable.

Terminology and mathematical conventions follow ISO/CIE standards where available. Non-standard terms are defined on first use.

3. DISCLOSED INNOVATIONS

3.1 Three-Layer Competency Framework for Colour Assessment

Problem addressed

Existing digital and physical colour-vision assessment instruments measure a single capability — discrimination — through ordering or threshold tasks. No existing instrument architecture distinguishes between perceptual discrimination (detection that a colour difference exists), analytical attribution (identification of which dimension changed), and linguistic communication (translation of the percept into standardised terminology) as separable and separately scored components of professional colour competency. The absence of this distinction leaves the instrument unable to differentiate between observer profiles that behave differently in professional practice: the observer who detects but cannot describe, the observer who describes but cannot judge magnitude, and the observer who does both reliably.

Prior art and distinction

Clinical colour-vision tests (FM100, Cambridge Colour Test, Colour Assessment and Diagnosis test, Farnsworth D15) measure discrimination only. The Japanese Colour Aptitude Test (JCAT; Hirschler, Gay & Ferreira de Oliveira, 2002) includes a Colour-Attribute Discrimination Test that requires observers to identify whether a colour difference is a change in hue, value or chroma; this is a physical-artefact precedent for attribution measurement but is not integrated with discrimination or communication measurement in a single assessment architecture, is not available as a digital platform, and does not produce dimension-resolved or layer-resolved scoring.

Neuropsychological evidence for separation of colour attribution from colour naming is published in the basic-science literature (e.g., Siuda-Krzywicka et al., 2019 on neural double dissociation; Skelton et al., 2017 on pre-verbal categorical perception; Hardy et al., 2005 on lifespan dissociation between discrimination and naming). This basic-science evidence establishes the empirical legitimacy of the three-layer distinction but has not been operationalised as an assessment architecture. No published assessment instrument known to the author integrates all three measurement layers.

The distinction claimed here is the operationalisation of the three-layer structure as an integrated assessment architecture: instruments, scoring model, and reporting format that treat discrimination, attribution, and communication as distinct measurable constructs with independent scoring pipelines and independent validation strategies.

Claimed subject matter (broad)

A computer-implemented method and system for professional colour competency assessment, characterised by:

- (a) measurement of an observer's ability to *detect* that a colour difference exists between two or more stimuli (the **discrimination layer**);
- (b) measurement, independently of (a), of the observer's ability to *identify* which perceptual dimension of the stimulus — hue, lightness, or chroma — has changed (the **attribution layer**);
- (c) measurement, independently of (a) and (b), of the observer's ability to *translate* the perceptual content of the stimulus into standardised linguistic or symbolic terminology, and to reconstruct a target stimulus from such a description (the **communication layer**);
- (d) independent scoring pipelines for layers (a), (b), and (c), producing three scalar or vector scores per observer per session that are reported separately and are not aggregated into a single summary without preservation of the separate layer scores;
- (e) a reporting format in which discrimination, attribution, and communication scores are individually visible to the administrator, the observer, and downstream systems (certification, training recommendation, qualification decision).

The broad claim covers any computer-implemented colour-assessment system that implements the three-layer measurement architecture as defined above, independently of the specific instruments used to measure each layer, the specific scoring formulas, the specific stimulus parameters, or the specific reporting interface.

Enabling description

An implementation of the three-layer framework requires:

1. **Stimulus generation infrastructure** capable of producing stimuli that vary exactly one perceptual dimension at a time, for discrimination and attribution tasks, and stimuli accompanied by structured linguistic descriptions for communication tasks. A perceptually uniform colour space such as OKLCH (Ottoosson, 2020) with per-hue gamut-aware trial construction (Section 3.7) is suitable but not the only suitable basis.
2. **Task structure for each layer:** - Discrimination tasks: ordering tasks, forced-choice comparison tasks, or adaptive-threshold tasks. The observer response is a spatial arrangement, a choice between

two or more alternatives, or a threshold detection. - Attribution tasks: presentation of stimulus pairs differing in exactly one dimension. The observer response is identification of the changed dimension (hue / lightness / chroma) and, optionally, the direction of change within that dimension. -

Communication tasks: presentation of a stimulus paired with candidate linguistic descriptors (colour-to-description direction), or presentation of a linguistic descriptor paired with candidate stimuli (description-to-colour direction). The observer response is selection from the candidate set.

3. Independent scoring pipelines. Each layer receives a dedicated scoring formula that produces a 0–100 scalar score or an equivalent. Scores may be further decomposed per dimension (hue, lightness, chroma) within each layer, producing a 3×3 matrix of scores (see Section 3.2).

4. Layer-preserving reporting. The session output includes at minimum three layer-level scores — discrimination, attribution, communication — that are individually visible and downstream-accessible. Aggregated summary scores may be produced in addition but do not replace the layer-level scores.

5. Layer-differentiated training module selection. An observer scoring high on discrimination and low on attribution is recommended analytical training; an observer scoring high on attribution and low on communication is recommended vocabulary training; and so on. The layer-resolved score structure supports targeted training-module selection that single-score instruments cannot provide.

Embodiments

A reference implementation of the three-layer framework is the ColorAptitude platform developed by the author, in which:

- The discrimination layer is implemented through a Hue Ordering Diagnostic (Section 3.4), Lightness Ordering Diagnostic, Chroma Ordering Diagnostic, and Chroma Threshold Diagnostic.
- The attribution layer is implemented through an Attribution Exercise with cardinal-nearest classification (Section 3.6 and Section 3.8).
- The communication layer is implemented through a Communication Exercise with a structured vocabulary of sixteen hue descriptors $\times$ three lightness levels $\times$ three chroma levels (144 distinguishable descriptions).
- Scores are reported as a 3×3 matrix (Section 3.2).

Alternative embodiments may use different instrument sets, different scoring formulas, different vocabulary structures, and different stimulus-generation pipelines. The broad claim is not limited to the reference implementation.

3.2 Three-by-Three Competency Matrix as Reporting Architecture

Problem addressed

A three-layer assessment architecture (Section 3.1) produces three layer-level scores, but it does not by itself indicate which of the three colour dimensions (hue, lightness, chroma) contributes to a given layer score. An observer may perform well on hue discrimination but poorly on chroma discrimination, or well on lightness attribution but poorly on hue attribution. Single-score or single-axis reporting loses this information. A reporting architecture that preserves dimensional and layer-wise structure simultaneously is required for targeted training recommendation and for fine-grained qualification decisions.

Prior art and distinction

Dimension-resolved scoring exists in the clinical literature for discrimination: the FM100 reports segmentwise errors that can be mapped to hue regions, and the Cambridge Colour Test reports thresholds on protan / deutan / tritan axes. These are dimension-resolved within the discrimination layer only. No known published instrument or patent reports scores across the joint space of three colour dimensions $\times$ three competency layers.

The claimed subject matter is the joint reporting architecture: simultaneous resolution of the competency score across both the dimensional axis and the layer axis, producing a structured 3×3 matrix that preserves cross-axis patterns.

Claimed subject matter (broad)

A computer-implemented method for reporting professional colour competency assessment results, characterised by:

- (a) decomposition of the session-level competency score along two orthogonal axes: a *dimension* axis with values {hue, lightness, chroma}, and a *layer* axis with values {discrimination, attribution, communication};
- (b) reporting of a score matrix with at least three rows and three columns, in which each cell (dimension, layer) contains an independent scalar score reflecting the observer's competency in that specific intersection;
- (c) marginal aggregation along either axis: row-wise aggregation produces per-dimension scores across layers; column-wise aggregation produces per-layer scores across dimensions;
- (d) retention of the cell-level structure in all downstream uses: training module selection, certification scope definition, and cross-session progress tracking operate on cell-level, not only on aggregated scores;
- (e) presentation of the matrix to human readers (observer, administrator, certifier) in a form that preserves the 3×3 structure — typically a tabular presentation in which cells are individually legible.

The broad claim covers any computer-implemented assessment reporting system that decomposes scores along both dimensional and competency-layer axes and preserves that decomposition in downstream use, independently of the specific instruments, scoring formulas, or visualisation choices.

Enabling description

Implementation of the 3×3 matrix reporting requires:

1. **Per-instrument dimension mapping.** Each assessment instrument declares which dimensions it measures. An ordering diagnostic that varies only hue (Section 3.4) maps its score to the hue row; an ordering diagnostic that varies only lightness maps to the lightness row. Attribution and Communication Exercises that test all three dimensions produce dimension-resolved scores for each dimension they cover.
2. **Per-instrument layer mapping.** Each assessment instrument declares which layer it measures. Ordering and threshold diagnostics map to the discrimination layer. Attribution Exercises map to the attribution layer. Communication Exercises map to the communication layer. Multi-layer instruments (rare but possible) may contribute to more than one cell.
3. **Cell-level score aggregation.** Within each cell (dimension, layer), scores from one or more instruments are aggregated by a defined rule — typically a time-weighted mean, a worst-case minimum, or an instrument-specific priority. The aggregation rule is explicit per cell.
4. **Marginal aggregation rules.** Row-wise aggregation (across layers for a given dimension) and column-wise aggregation (across dimensions for a given layer) produce the six marginal scores. A single overall score — matrix sum or mean — may be computed but does not replace the cell or marginal scores in reporting.
5. **Downstream cell-awareness.** Training recommendation consults the cell-level scores, not only marginal or aggregate scores. Certification decisions may refer to cell-level thresholds in addition to overall thresholds.
6. **Preservation of partial matrices.** A session that administers only a subset of instruments produces a partial matrix in which uncovered cells are marked as not-measured rather than being filled with imputed or default values. Marginal aggregates are computed only over measured cells.

Embodiments

The reference implementation in ColorAptitude produces the following 3×3 matrix per session:

Dimension / Layer	Discrimination	Attribution	Communication
Hue	Hue Ordering Diagnostic	Attribution Exercise (hue trials)	Communication Exercise (hue naming)

Dimension / Layer	Discrimination	Attribution	Communication
Lightness	Lightness Ordering Diagnostic	Attribution Exercise (lightness trials)	Communication Exercise (lightness naming)
Chroma	Chroma Ordering Diagnostic + Chroma Threshold Diagnostic	Attribution Exercise (chroma trials)	Communication Exercise (chroma naming)

Each cell contains a 0–100 scalar score plus optional confidence interval. Row marginals (e.g., overall hue competency across all three layers) and column marginals (e.g., overall discrimination across all three dimensions) are reported alongside cell scores.

Alternative embodiments may use a larger matrix (for example, 4×3 with a magnitude-scaling dimension added, or 3×4 with a consistency layer added), may redefine cells, or may use different aggregation rules. The broad claim requires only the joint decomposition along dimensional and competency-layer axes and the preservation of cell-level structure in reporting.

3.3 Differentiated Three-Track Validation Strategy

Problem addressed

A three-layer competency framework (Section 3.1) raises a distinct validation problem: the three layers are not at the same stage of psychometric development. The discrimination layer admits classical norm-referenced validation against existing population data (FM100 normative literature). The attribution layer admits known-groups validation if a trained-versus-untrained contrast can be established. The communication layer, in many industrial contexts, cannot admit either form of validation at the scale required, because a coherent trained expert population does not yet exist in the general practitioner population. Applying a single validation strategy uniformly across the three layers is either over-claiming (presenting communication as if it had discrimination-layer rigour) or under-claiming (restricting the instrument to discrimination only and discarding the wider framework). A differentiated validation strategy per layer is required.

Prior art and distinction

Classical psychometric validity theory (e.g., Messick's framework; AERA, APA, NCME *Standards for Educational and Psychological Testing*, 2014) recognises multiple types of validity — criterion, construct, content, known-groups — and acknowledges that new constructs pass through construct-validity stages before admitting classical norm-referenced validation. Selection of the appropriate validity type for a given construct is a standard psychometric practice.

The distinction claimed here is the explicit operationalisation of a *differentiated* validation strategy that applies *different* validity types to *different* measurement layers *within a single assessment instrument*. The strategy treats the layers asymmetrically by design, corresponding to their distinct psychometric development stages, and publishes validation claims with differentiated scope per layer rather than applying a uniform claim across the instrument. No known published validation strategy for a multi-layer colour-competency assessment instrument uses this differentiated approach.

Claimed subject matter (broad)

A validation methodology for a multi-layer colour-competency assessment instrument, characterised by:

- (a) classification of each measurement layer into a psychometric validity stage appropriate to the empirical maturity of the construct measured by that layer;
- (b) application of a distinct validation protocol per layer: criterion-referenced validation for layers with established population norms; known-groups validation for layers with an identifiable trained-versus-untrained contrast; construct-validity monitoring for layers whose target population is still in development;
- (c) differentiated reporting of validation status per layer in the instrument's public documentation, including the permissible claim scope for each layer under its current validation stage;
- (d) an explicit longitudinal path by which each layer progresses from its current validation stage toward higher-order validation as empirical data accumulate through curriculum deployment, cohort development, and planned data-collection campaigns;
- (e) published deferral of norm-referenced claims for layers that have not yet reached the necessary empirical maturity, with explicit `unrated` or equivalent status in operational scoring output.

Enabling description

Implementation of the differentiated validation strategy requires:

1. **Per-layer validation stage classification.** Each measurement layer is classified, by the instrument authors, into one of: criterion-validity stage (norm-referenced validation is feasible and planned); known-groups-validity stage (trained-versus-untrained contrast is feasible and planned); construct-validity stage (target population still in development, norm-referenced validation not currently feasible).
2. **Per-layer validation protocol.** For each stage, a specific protocol is published:
 - Criterion stage: sample-size target, stratification criteria, two-run or equivalent protocol, cross-instrument validation against established physical instruments, band-threshold derivation from empirical population distribution.
 - Known-groups stage: recruitment criteria for the "trained" group and the contrast "untrained" group, minimum sample size per group, statistical test for group separation, ROC-derived

threshold for optimal discrimination between groups. - Construct stage: longitudinal cohort development plan, interim consistency measures (within-observer stability, curriculum-alignment indicators), explicit deferral of population-referenced claims.

3. Differentiated scope positioning in all public communication. The instrument's scoring output, reports, certification documents, and scientific publications label each score with its current validation-stage scope. A score at construct-validity stage is reported with an explicit provisional indicator (for example, `unrated` status) that is preserved through all downstream display and interpretation.

4. Longitudinal validation path. The instrument's public documentation describes the planned progression of each layer from its current stage toward higher-order validation, including the specific data-collection campaigns and infrastructure milestones that enable progression.

5. Independence of layer-level validation claims. Progression of one layer does not imply progression of another. The instrument may achieve criterion-validity on discrimination while remaining at construct-validity on communication, and the public claim scope reflects this asymmetry at each stage.

Embodiments

The reference implementation in ColorAptitude applies the differentiated strategy as follows:

- **Discrimination layer — criterion-validity stage.** Planned Track 1 data collection through a sampling mailing to existing Nederlandse Kleurenschool and SNKI customers, target sample $N \geq 200$ in Dutch-EU first phase, two-run protocol with minimum seven-day interval, Tier 2 hardware calibration mandatory, age stratification across six bins, gender balance per bin. Cross-instrument validation against physical FM100 administration planned as Fase V separate study.
- **Attribution layer — known-groups-validity stage.** Planned pilot expert-panel study, target $N = 15\text{--}25$ trained participants (Kleurenschool instructors, SNKI-certified practitioners, industrial colour-quality managers with documented analytical colour-judgment experience), contrasted against matched subset of Track 1 participants without formal colour-analytical training. ROC-derived band thresholds planned.
- **Communication layer — construct-validity stage.** Deferred population-referenced validation. Longitudinal cohort development through Nederlandse Kleurenschool graduate cohorts. In the current release, communication-layer scores are reported as within-observer consistency and curriculum-alignment indicators. `unrated` status preserved in all downstream display.

Alternative embodiments may use different recruitment strategies, different sample-size targets, or different contrast groups. The broad claim requires only the differentiated per-layer strategy and the preservation of differentiated claim scope in public communication.

3.4 Five-Segment 18-Cap Hue Ordering Diagnostic with Three Red-Green Opponent Segments

Problem addressed

The Farnsworth-Munsell 100 Hue Test (FM100; Farnsworth, 1957) organises 85 caps into four trays, producing only two segments that cross the red-green opponent axis. This limits the test's sensitivity to deutan and protan confusion patterns, because a CVD observer's errors may be distributed across segments in ways that are not individually diagnostic. A segmentation that provides additional red-green opponent segments, and that uses a perceptually grounded five-principal-hues organisation rather than the four-unique-hues organisation implicit in the FM100, improves confusion-line detection without substantially increasing test duration.

Prior art and distinction

The FM100 uses a four-segment structure. Its scoring is based on the total error score (TES) formula: for each movable cap, $\text{partial error} = |\text{index}(\text{prev}) - \text{index}(\text{placed})| + |\text{index}(\text{next}) - \text{index}(\text{placed})| - 2$. The TES formula is prior art of Farnsworth (1957).

Fairchild (2018) distinguishes between four unique hues (Hering: red, yellow, green, blue) and five principal hues (Munsell: red, yellow, green, blue, purple), and argues that discrimination tasks correlate more strongly with the five-principal organisation. This is perceptual-science prior art but has not been implemented in a commercial discrimination assessment instrument to the author's knowledge.

The distinction claimed here is the specific integration of the five-principal-hues organisation with an 18-cap per-segment structure that yields three red-green opponent segments (red-purple, blue-green, green-yellow), implemented digitally with OKLCH gamut-aware cap generation (Section 3.7) and analytical worst-case TES normalisation.

Claimed subject matter

Broad claim. A colour-discrimination ordering diagnostic characterised by organisation of the stimulus set into five perceptually defined segments spanning the hue circle through the five principal hues (red, yellow, green, blue, purple), with the segments chosen such that at least three of the five contain red-green opponent transitions. The broad claim covers any ordering-based hue-discrimination instrument with this five-segment three-opponent organisation, independently of the specific cap count per segment, the specific cap size, or the specific stimulus medium (physical chips, digital screen, VR).

Narrow claim. The specific implementation in which:

- Each segment contains 18 caps, of which 17 are movable and one is a fixed anchor at the segment boundary;

- The five segments together produce a sequence of 90 caps with 5 fixed anchors and 85 movable caps;
- The five segments are named and arranged as: red-purple, blue-green, green-yellow, yellow-red, and one residual segment covering the remaining hue region between the principal hues;
- The error scoring uses the FM100 TES formula applied per movable cap and summed across all five segments;
- The worst-case TES is analytically derived as 288 per segment and 1440 across the five-segment diagnostic;
- The TES-like scoring formula $\text{tesLikeScore} = (1 - \text{clamp}(\text{totalError} / 1440, 0, 1)) \times 100$ produces a 0–100 score in which higher indicates better performance;
- An optional cross-reference rescaling factor of $1440 / 1584 \approx 0.91$ is applied for comparison against FM100 normative literature;
- Three of the five segments (red-purple, blue-green, green-yellow) contain red-green opponent transitions, providing three independent observations of red-green confusion-line patterns rather than the two observations provided by the FM100's four-segment structure.

Enabling description

1. **Segment boundary determination.** The five principal hue anchors are located at the following approximate OKLCH hue angles (derived from Munsell principal hues via the Munsell-to-OKLCH transformation): - Red: $\approx 29^\circ$ - Yellow: $\approx 110^\circ$ - Green: $\approx 143^\circ$ - Blue: $\approx 264^\circ$ - Purple: $\approx 328^\circ$

The segment boundaries are placed at the midpoints between adjacent principal hues, producing five arcs that each cover approximately 72° of the hue circle but with non-uniform widths reflecting the perceptual non-uniformity of the OKLCH hue space.

2. **Cap placement within segment.** Within each 72° -arc segment, 18 caps are placed at perceptually equal spacing along the OKLCH hue dimension. The first cap is fixed as the segment anchor; the remaining 17 caps are shuffled and presented as movable caps.

3. **Constant lightness and chroma.** All 90 caps across all five segments use $L = 0.70$ in OKLCH and $C = 0.95 \times \text{minimum in-gamut chroma}$ across all hue positions. The minimum in-gamut chroma is computed by binary search (20 iterations, precision ≈ 0.0004 chroma units) ensuring that no cap is clipped by the sRGB or P3 gamut at any hue position (Section 3.7).

4. **Observer task.** The observer arranges the 17 movable caps in each segment into a continuous perceptual hue sequence. The task may be performed segment-by-segment or across the full 85-cap sequence depending on UI design. The reference implementation presents segments sequentially to preserve comparability with the FM100's tray-by-tray administration.

5. Scoring. For each movable cap at position placed, with neighbours at positions prev and next in the observer's arrangement, partial error = $|\text{index}(\text{prev}) - \text{index}(\text{placed})| + |\text{index}(\text{next}) - \text{index}(\text{placed})| - 2$, where $\text{index}(k)$ denotes the correct ordinal position of cap k in the ground-truth sequence. Total error (TES) is the sum of partial errors across all 85 movable caps. The 0–100 score is $\text{tesLikeScore} = (1 - \text{clamp}(\text{TES} / 1440, 0, 1)) \times 100$.

6. Red-green confusion detection. For each of the three red-green opponent segments (red-purple, blue-green, green-yellow), the segment TES is computed separately. An observer with elevated segment TES in all three red-green segments simultaneously, relative to segment TES in the two non-opponent segments (yellow-red, residual), is flagged as showing a red-green confusion pattern consistent with deutan or protan tendencies. The specific diagnostic thresholds depend on population-norm data (Section 3.3).

7. Worst-case TES derivation. The analytical worst-case TES per 18-cap segment is derived as follows. For a sequence of n movable caps with partial-error formula as above, the maximum total error occurs when the arrangement is a complete reversal of the correct sequence. For $n = 17$ movable caps (one fixed anchor plus 17 movable in an 18-cap segment), $\text{max TES} = 288$. Across five segments: $5 \times 288 = 1440$.

Embodiments

The reference implementation in ColorAptitude presents segments sequentially with an option for the observer to pause between segments, includes five unscored warm-up caps at the start of each segment (ASTM E1499-16 §7.1.1), and caps total session duration at 20 minutes (ASTM E1499-16 §8.1.1).

Alternative embodiments may include: different per-segment cap counts (e.g., 12 or 24 caps per segment); different stimulus media (physical chips produced as a colour-ordering set; VR or AR presentation); different cap shapes; different per-segment lightness or chroma anchors (the broad claim is not limited to $L = 0.70$); different screen technologies (LCD, OLED, e-paper); or different diagnostic threshold schemes for CVD detection. The broad claim does not require any of these specific embodiments.

3.5 Two-Path Display Calibration Architecture with ISO 3664:2009 P2 Compliance Logging

Problem addressed

Digital colour assessment requires display calibration to produce valid stimuli. Two deployment contexts exist with different calibration requirements:

- (i) *home or workplace deployment* on observer-owned hardware without hardware colorimeters, where the assessment is used for training, indicative evaluation, or preliminary screening; and

(ii) *compliance-oriented deployment* for formal observer qualification under ASTM E1499-16 §5.3, where rigorous display characterisation is required.

A single-path calibration protocol either fails deployment context (i) — if it requires hardware instrumentation — or fails deployment context (ii) — if it lacks objective measurement. A two-path architecture addresses both contexts with appropriate rigour in each, and documents viewing conditions at a level of detail sufficient for normative analysis.

Prior art and distinction

Software-based display calibration wizards that rely on observer self-report are common in consumer-grade software (for example, gamma adjustment sliders in operating systems). Hardware colorimeter calibration software from vendors including Datacolor (SpyderPro, SpyderX) and X-Rite (i1Display) is commercially available. ISO 3664:2009 *Graphic technology and photography — Viewing conditions* specifies viewing-condition requirements for softproof evaluation, including the P2 condition ambient-to-display luminance ratio.

The distinction claimed here is the specific integrated architecture:

- a software-only calibration path (Path A) suitable for observer-managed deployment, with a seven-step validation protocol targeting distinct display characteristics;
- a hardware-measured path (Path B) integrated with commercial colorimeter reports through structured parsing, with automated computation of ISO 3664:2009 P2 compliance ratio and mixed-chromatic-adaptation risk flagging;
- a unified composite scoring system that accommodates both paths with revised weighting, preserving cross-path comparability;
- session-validity logic that combines calibration band, temporal freshness, and session-guard outcomes into a single validity classification;
- certificate-tier annotation that records the calibration path used for each formal qualification session.

No known published calibration architecture known to the author combines all of these elements.

Claimed subject matter

Broad claim. A display calibration architecture for digital colour assessment systems, characterised by:

- (a) provision of a *software-only* calibration path executable by the observer without hardware instrumentation, producing a composite calibration score bounded by the limit of unaided validation;
- (b) provision of a *hardware-measured* calibration path integrated with commercial colorimeter reports, producing a composite calibration score capable of reaching the maximum validation band;

- (c) automated parsing of hardware-calibration reports to extract gamma, white-point chromaticity, black- and white-luminance, and gamut-coverage metrics for storage in session metadata;
- (d) automated computation of an ambient-to-display luminance ratio per ISO 3664:2009 P2 specification, producing a per-session compliance status (mandatory threshold, recommended threshold, non-compliant) stored in session metadata;
- (e) automated computation of a mixed-chromatic-adaptation risk flag based on the difference between ambient correlated colour temperature and display white-point correlated colour temperature;
- (f) unified composite scoring that weights Path A subjective validations and Path B objective measurements differently but produces comparable band classifications across paths;
- (g) certificate-tier recording: each formal qualification certificate records the calibration path used for the qualifying session, enabling downstream verifiers to interpret the qualification context.

Narrow claim. The specific implementation in which:

- Path A comprises seven validation steps: gamma (20% weight), black level (10%), white level (10%), luminance (15%), white point (15%), gamut (20%), neutrality (10%);
- Path A maximum composite band is capped at Good (≥ 65) to reflect the residual subjectivity of unaided chromatic validations;
- Path B data override Path A on the gamma, white-point, and gamut steps, with reduced weight for the remaining subjective steps;
- Path B composite band may reach Excellent (≥ 85);
- Calibration records are considered current for 30 days; older records trigger an expired flag;
- Band classification is: Excellent (≥ 85), Good (≥ 65), Fair (≥ 40), Poor (< 40);
- Session validity logic: invalid if band = Poor; reduced if band = Fair or calibration expired; full otherwise;
- ISO 3664 P2 compliance: mandatory threshold at ambient-to-display luminance ratio $\leq 1/4$, recommended threshold at ratio $\leq 1/8$;
- Mixed-adaptation flag triggers at CCT delta exceeding 2500 K between ambient and display.

Enabling description

1. Path A seven-step wizard.

Step 1 — Gamma. A grey-step matching task presents a striped alternating black-and-white patch next to a solid grey patch. The observer adjusts a slider until the solid patch appears equal in lightness to the striped patch. The slider position indexes the display gamma; pass criterion is slider within ± 0.3 of $\gamma = 2.2$.

Step 2 — Black level. Five dark squares at luminance levels corresponding to 1%, 2%, 3%, 4%, and 5% of display maximum are presented against black background. Pass criterion is observer reports visibility of at least 3 of 5 squares.

Step 3 — White level. Five near-white squares at 95%, 96%, 97%, 98%, 99% of display maximum are presented against white background. Pass criterion is observer reports visibility of at least 3 of 5 squares.

Step 4 — Luminance. An 18% grey patch is presented against white background. Pass criterion is observer reports the patch is clearly visible with distinct boundary.

Step 5 — White point. Two near-white patches at 6000 K (warm) and 7500 K (cool) are presented side by side. Pass criterion is observer reports no dominant colour cast on either patch.

Step 6 — Gamut. Three pairs of high-chroma OKLCH stimuli in red, green, and blue regions are presented. Pass criterion is observer reports all three pairs as discriminable.

Step 7 — Neutrality. Six grey fields at different lightness levels are presented. Pass criterion is observer reports at most 1 of 6 fields as having a colour cast.

2. Path B integration. The observer runs a commercial colorimeter calibration (reference implementation: Datacolor SpyderPro) and exports the calibration report as a PDF. The ColorAptitude platform parses the report with a structured extractor that locates and extracts: gamma value, white-point chromaticity (x, y or u', v'), white-point ΔE from target, grey-series ΔE , black luminance (cd/m^2), white luminance (cd/m^2), and gamut coverage percentage (sRGB, and optionally P3). Extracted values are written to the session metadata record.

3. ISO 3664 P2 compliance computation. The observer reports the ambient illuminance at the display face on a white reference surface, either via an automated reading from the colorimeter's ambient sensor where available, or via manual entry from a separate lux meter. The platform computes the ratio = ambient illuminance (lx, converted to cd/m^2 via the reference surface BRDF) / display white luminance (cd/m^2). Classification: mandatory compliant if ratio $\leq 1/4$; recommended compliant if ratio $\leq 1/8$; non-compliant if ratio $> 1/4$. Stored in `Session.metadata.iso3664P2Status`.

4. Mixed-adaptation flag. Ambient CCT is reported (from colorimeter ambient sensor or manual entry). Display white-point CCT is taken from the Path B gamma / white-point measurement, defaulting to 6500 K if Path B is not available. CCT delta = $|\text{ambient CCT} - \text{display CCT}|$. Flag triggers if delta > 2500 K. Stored in `Session.metadata.mixedAdaptationRisk`.

5. Composite score formulas.

Path A only: $\text{compositeScore} = 0.20 \times \text{gammaScore} + 0.10 \times \text{blackLevelScore} + 0.10 \times \text{whiteLevelScore} + 0.15 \times \text{luminanceScore} + 0.15 \times \text{whitePointScore} + 0.20 \times \text{gamutScore} + 0.10 \times \text{neutralityScore}$; each component score $\in \{0, 100\}$.

Path B available: Path B measurements produce objective scores for gamma, white-point, and gamut. These override the corresponding Path A components. The remaining Path A components (black level, white level, luminance, neutrality) retain their original weights but contribute at reduced effective weight of 0.05, 0.05, 0.08, 0.05 respectively, with the remainder of the weight allocated to Path B components in proportion to their original weights.

6. Session validity logic. Implemented as a pure function of `calibrationBand`, `calibrationExpired` (days since calibration > 30), and `calibrationFailed` (`calibrationBand` = Poor):
 - If `calibrationFailed` → validity = invalid.
 - Else if `calibrationExpired` OR `calibrationBand` = Fair → validity = reduced.
 - Else → validity = full.

7. Certificate-tier recording. Every formal qualification certificate includes a field `calibrationTier` $\in$ {Tier1, Tier2}. Tier1 indicates Path A only; Tier2 indicates Path B available and used. Qualification under ASTM E1499-16 §5.3 requires Tier2.

Embodiments

Reference implementation uses the Datacolor SpyderPro as the Path B reference instrument, with structured parser targeting the SpyderPro PDF report format. Alternative embodiments may target other colorimeters (X-Rite i1Display, Calibrite, Klein, Minolta CA-310) with appropriate parser updates. The broad claim is not limited to any specific colorimeter brand.

The seven-step Path A wizard is one specific realisation; alternative embodiments may use more or fewer steps, different weighting, or different per-step pass criteria. The broad claim requires only the separation into two paths, the ISO 3664 P2 ratio computation, the mixed-adaptation flag, and the certificate-tier recording.

3.6 Cardinal-Nearest Hue Classification for Attribution Scoring

Problem addressed

Attribution-layer scoring (Section 3.1) requires a method for classifying an observer's reported hue-direction response to a stimulus pair. Two existing approaches present difficulties: free unique-hue matching (where the observer identifies a stimulus that appears as "pure" red, green, blue, or yellow) is subject to the circularity objection raised by Logvinenko (2012), because the unique-hue stimuli used as reference are themselves defined by observer self-report; and *a/b* axis decomposition in CIELAB produces dual-label ambiguity, because any hue shift can be decomposed onto orthogonal *a* and *b* components, both of which have direction signs that do not map cleanly onto a single perceptual-direction label.

A classification method that avoids both the circularity objection and the dual-label ambiguity, while producing a single, empirically grounded, reproducible direction label per trial, is required for valid attribution-layer scoring.

Prior art and distinction

Unique-hue measurement and matching have a long history in colour science. Kuehni (2008) reports mean-observer hue angles from 102 observers with two repetitions each: red at Munsell hue angle 16.5° , yellow at 78.7° , green at 157.7° , blue at 220.3° . Kuehni (2014) discusses systematic differences between these empirical mean-observer angles and the CIECAM02 default unique-hue angles, which are derived from Swedish NCS anchoring. Logvinenko (2012) proposes partial hue-matching from a closed stimulus set as an alternative to free unique-hue matching, to avoid the circularity of observer-defined references.

The distinction claimed here is the specific application: use of empirical mean-observer cardinal hue angles (Kuehni 2008) as fixed sector boundaries for a closed-set classification of an observer's hue-direction response, in the context of an attribution-layer assessment scoring pipeline. The cardinal sectors function as a closed reference set in Logvinenko's sense, and the empirical angles ensure that the classification is grounded in population data rather than in individual self-report. The combined effect is an attribution-scoring method that is simultaneously empirically grounded (via Kuehni) and methodologically sound (via Logvinenko's closed-set argument).

Claimed subject matter

Broad claim. A method for classifying an observer's hue-direction response in a colour-attribution task, characterised by:

- (a) definition of four cardinal hue directions corresponding to red, yellow, green, and blue, with each cardinal direction anchored at an empirically derived mean-observer hue angle;
- (b) partitioning of the OKLCH hue circle (or equivalent perceptually uniform hue-angle space) into four sectors, each sector containing exactly one cardinal anchor;
- (c) classification of the observer's reported direction by identification of which cardinal sector contains the angular shift between reference and comparison hue;
- (d) scoring of the observer's response as correct if the classified cardinal direction matches the ground-truth direction determined by the sign of the angular shift from reference to comparison in the stimulus construction, and incorrect otherwise;
- (e) use of a closed response set (four cardinal alternatives) rather than free unique-hue matching, to eliminate the circularity inherent in observer-defined reference stimuli.

Narrow claim. The specific implementation in which:

- The four cardinal anchors are placed at Munsell hue angles 16.5° (red), 78.7° (yellow), 157.7° (green), and 220.3° (blue), following Kuehni (2008);
- Munsell hue angles are transformed to OKLCH hue angles via the Munsell-to-OKLCH chromaticity mapping used throughout the reference implementation;

- Sector boundaries are placed at the angular midpoints between adjacent cardinal anchors;
- The observer's response is collected through a forced-choice UI with four options labelled "reddish", "yellowish", "greenish", "bluish", presented in consistent spatial order across trials;
- Direction-labelling confusions (e.g., observer reports "yellowish" for a stimulus in the green sector) are scored as incorrect and additionally logged for diagnostic analysis as confusion-matrix entries.

Enabling description

1. Cardinal anchor definition. The four cardinal anchors are fixed as the empirical mean-observer values from Kuehni (2008):

Cardinal	Munsell hue angle	OKLCH hue angle (approximate)
Red	16.5°	≈ 29°
Yellow	78.7°	≈ 110°
Green	157.7°	≈ 143°
Blue	220.3°	≈ 264°

The OKLCH values are derived by the Munsell-to-OKLCH transformation used throughout the ColorAptitude stimulus generation pipeline and may differ slightly in alternative implementations using different transformation chains. The claim is indifferent to the choice of perceptually uniform hue-angle space; the cardinal anchors are specified in the space of implementation.

2. Sector boundaries. Boundaries are placed at the angular midpoints between adjacent cardinals in the chosen hue-angle space. In OKLCH (approximate values): - Red/Yellow boundary: ≈ 70° - Yellow/Green boundary: ≈ 127° - Green/Blue boundary: ≈ 204° - Blue/Red boundary: ≈ 329°

3. Stimulus construction. Each attribution trial presents a reference stimulus at OKLCH (L, C, H_ref) and a comparison stimulus at OKLCH (L, C, H_ref + Δh), with L and C held constant across the two stimuli and Δh drawn from a range that produces supra-threshold hue discrimination without crossing a sector boundary (typically Δh ∈ [10°, 40°] depending on the trial's target difficulty). The ground-truth direction is defined as the cardinal sector in which the angular shift from H_ref to H_ref + Δh falls, relative to the reference hue.

4. Observer response collection. A four-alternative forced-choice UI presents the four cardinal labels. The observer selects the label that best describes the direction of the comparison from the reference.

5. Scoring. Let gt_cardinal ∈ {red, yellow, green, blue} denote the ground-truth cardinal, and obs_cardinal ∈ {red, yellow, green, blue} denote the observer's response. The trial is scored as correct if gt_cardinal = obs_cardinal, and incorrect otherwise.

6. Confusion-matrix logging. Across the trial set, incorrect responses are aggregated into a 4×4 confusion matrix indexed by (gt_cardinal, obs_cardinal). Specific confusion patterns — for example, elevated red-green confusion rates — are logged and reported as diagnostic indicators of attribution-layer difficulty in specific perceptual regions.

7. Avoidance of Logvinenko circularity. The observer at no point identifies a stimulus as "pure red" or as "the observer's own unique red". The four cardinal anchors are fixed reference points against which the observer's classification is scored. The observer's task is a classification decision within a closed set of four alternatives, not a free unique-hue match. This structural property of the task avoids the circularity objection (Logvinenko, 2012) to unique-hue-based scoring.

Embodiments

The reference implementation presents the four cardinal labels in Dutch and English bilingual UI ("reddish / roodachtig", etc.), with consistent spatial ordering across trials to minimise position bias.

Alternative embodiments may use: different cardinal-anchor angle values (for example, population-specific means for non-Dutch-EU populations); extended cardinal sets (six or eight sectors for finer-grained classification); different UI label sets (e.g., numerical direction labels rather than colour-name labels); or implementation in different perceptually uniform colour spaces (CIELAB, CIECAM02, Oklab). The broad claim is not limited to any of these embodiments.

3.7 OKLCH Gamut-Aware Per-Hue Trial Generation with Binary-Search Safe Chroma

Problem addressed

A trial-generation pipeline for an attribution or discrimination task must produce stimulus pairs that vary exactly one perceptual dimension. If the requested chroma at a given (lightness, hue) combination exceeds the gamut limit of the target display (typically sRGB or P3), the rendering pipeline will clip the chromaticity, producing a stimulus that differs from the intended OKLCH specification. This clipping introduces uncontrolled cross-dimensional variation, invalidating the trial as a measurement of the intended dimension. A trial-generation method that guarantees in-gamut stimuli at all trial positions, with a safety margin to accommodate small rendering-pipeline rounding errors, is required for valid per-dimension measurement.

Prior art and distinction

OKLCH is prior art of Ottosson (2020) and is not claimed. Gamut mapping and gamut clipping are standard topics in colour imaging, with extensive prior art (Wyszecki & Stiles, 1982; Hunt, 2004). Per-hue gamut boundary computation is conceptually standard and is not claimed in isolation.

The distinction claimed here is the specific combined method: binary-search computation of the minimum in-gamut chroma across the full trial hue range, with a defined safety margin, applied as a pre-generation constraint on stimulus construction, such that all trials within a given assessment instrument share a single chroma value that is guaranteed in-gamut at every hue position used by the instrument.

Claimed subject matter (narrow)

A method for constructing stimuli for a single-dimension colour assessment trial in OKLCH, characterised by:

- (a) determination of the set of hue angles $H = \{h_1, h_2, \dots, h_n\}$ used by the assessment instrument;
- (b) for each $h_i \in H$, determination of the maximum in-gamut chroma C_i^{max} at (L, h_i) for the target display gamut (sRGB or P3), by binary search between 0 and a theoretical upper bound, with termination criterion defined by a precision threshold (reference implementation: 20 iterations, precision ≈ 0.0004 chroma units);
- (c) selection of the applied chroma $C_{\text{apply}} = 0.95 \times \min(C_i^{\text{max}})$ across all $h_i \in H$, providing a 5% safety margin for rendering-pipeline rounding errors;
- (d) generation of all trial stimuli at OKLCH $(L, C_{\text{apply}}, h_i)$, for any $h_i \in H$ used in the trial, with assurance that no stimulus requires gamut clipping;
- (e) extension of the method to multi-segment hue-discrimination instruments: each segment's stimulus set uses its own segment-wide C_{apply} value derived from the segment's hue range, ensuring per-segment gamut safety even when different segments use different constant chromas.

Enabling description

1. **Binary-search procedure.** For target (L, h_i) , initialise $C_{\text{low}} = 0$, $C_{\text{high}} = 0.4$ (a chroma value known to exceed any sRGB or P3 gamut at any hue). For 20 iterations: compute $C_{\text{mid}} = (C_{\text{low}} + C_{\text{high}}) / 2$; transform OKLCH (L, C_{mid}, h_i) to target display RGB via the standard OKLCH $\rightarrow$ linear-sRGB-or-P3 $\rightarrow$ gamma-corrected RGB transformation chain; if any RGB component is outside $[0, 1]$, set $C_{\text{high}} = C_{\text{mid}}$; otherwise set $C_{\text{low}} = C_{\text{mid}}$. After 20 iterations, $C_i^{\text{max}} = C_{\text{low}}$.
2. **Hue set enumeration.** For a Hue Ordering Diagnostic (Section 3.4), H is the set of 90 hue angles used across the five segments. For an Attribution Exercise, H is the set of hue angles used as reference stimuli. For a Chroma Ordering Diagnostic, H is a single hue per tray.
3. **Applied chroma computation.** $C_{\text{apply}} = 0.95 \times \min_i C_i^{\text{max}}$. The 5% safety margin accommodates: floating-point rounding in the OKLCH $\rightarrow$ RGB transformation; sub-pixel rendering variation between display technologies; and small inter-display variation in observed gamut coverage at specification boundaries.
4. **Per-segment variant.** For multi-segment instruments, each segment computes its own C_{apply} from its own hue subset. Segments with narrower hue ranges may have higher C_{apply} values; the Lightness Ordering Diagnostic with mixed-hue segments uses the minimum across the full lightness range as well as across hue, producing a segment-constant chroma that is in-gamut at all (L, h) combinations used in the segment.

5. Verification. Before each trial is presented, the constructed OKLCH (L, C_{apply}, h) value is transformed to target-display RGB as a sanity check; if any RGB component falls outside [0, 1], the trial is rejected and regenerated. This verification is a safety mechanism for floating-point edge cases; with the 5% safety margin applied correctly, rejection events should not occur.

Embodiments

The reference implementation uses sRGB as the default target gamut, with P3 as an optional extension for wide-gamut displays. The Munsell-to-OKLCH transformation is used where Munsell anchor angles are specified (Sections 3.4 and 3.6); this transformation is computed once per anchor set and cached.

Alternative embodiments may: target different display gamuts (Adobe RGB, BT.2020, etc.); use different safety margins (the broad claim is not limited to 5%); or use different search algorithms (bisection by Newton's method, linear search with fine quantisation). The narrow claim of this section is the combined method: binary-search maximum-chroma computation with a defined safety margin applied as a pre-generation constraint, guaranteeing in-gamut stimuli across the full trial hue range.

3.8 Kuehni-Anchored Closed-Set Attribution Scoring

Problem addressed

Section 3.6 establishes the cardinal-nearest classification method and its advantages over free unique-hue matching (Logvinenko circularity) and *a/b* decomposition (dual-label ambiguity). A complementary question is which specific empirical values to use as the cardinal-anchor hue angles. The default unique-hue angles used in CIECAM02 are derived from Swedish NCS anchoring. Kuehni (2014) shows that these values differ systematically from the cross-population empirical mean, and that for cross-population observer scoring the empirical mean is the more appropriate reference. Application of the Kuehni (2008) mean-observer values, specifically in the context of attribution-layer scoring within a three-layer competency assessment, is disclosed here.

Prior art and distinction

Kuehni (2008) publishes the mean-observer unique-hue angle data from 102 observers: red 16.5°, yellow 78.7°, green 157.7°, blue 220.3° (all Munsell hue angles). Kuehni (2014) reviews the state of the art and compares these values against CIECAM02 defaults. This prior art is public and is not claimed.

The distinction claimed here is the specific application: use of the Kuehni (2008) mean-observer values, rather than CIECAM02 default unique-hue angles, as the cardinal anchors in the closed-set attribution classification method described in Section 3.6. The claim is narrow because both the Kuehni data and the closed-set method rest on prior-art foundations; what is new is the specific combination, applied within an attribution-layer assessment scoring pipeline for professional colour competency.

Claimed subject matter (narrow)

A scoring method for a colour-attribution assessment task, characterised by:

- (a) adoption of four cardinal anchors at Munsell hue angles 16.5° (red), 78.7° (yellow), 157.7° (green), and 220.3° (blue), as published by Kuehni (2008);
- (b) transformation of these Munsell anchor values to the perceptually uniform colour space used for stimulus generation (OKLCH in the reference implementation), via the Munsell-to-target-space transformation used throughout the instrument;
- (c) use of the transformed anchors as the cardinal reference points in a closed-set hue-direction classification (per Section 3.6);
- (d) explicit rejection of the CIECAM02 default unique-hue angles as cardinal anchors in the same scoring pipeline, on the methodological ground that the CIECAM02 defaults are not representative of the cross-population observer mean;
- (e) documentation of the anchor provenance in the instrument's public specification, with citation to Kuehni (2008) and to Kuehni (2014) for the justification of the choice.

Enabling description

1. **Anchor values and source.** The four Munsell hue angles are 16.5° , 78.7° , 157.7° , and 220.3° , taken verbatim from Kuehni (2008), Table I, "Hinks et al. Average" column. Each value is the arithmetic mean across 102 observers, each measured with two repetitions separated by at least 24 hours.
2. **Transformation to stimulus space.** In the reference implementation, Munsell hue angles are transformed to OKLCH hue angles via a chained transformation: Munsell $\rightarrow$ CIE xyY (using Munsell Renotation data) $\rightarrow$ CIE XYZ $\rightarrow$ linear sRGB $\rightarrow$ Oklab $\rightarrow$ OKLCH. The transformation is computed once at platform startup and cached. Approximate OKLCH hue values for the four cardinal anchors are red $\approx 29^\circ$, yellow $\approx 110^\circ$, green $\approx 143^\circ$, blue $\approx 264^\circ$ (subject to the choice of intermediate transformation parameters).
3. **Verification against Kuehni (2014).** Kuehni (2014) publishes a comparison table between CIECAM02 defaults and the 102-observer empirical mean. The reference implementation has verified that its applied anchor values reproduce, within Munsell-to-OKLCH transformation tolerance, the values published in Kuehni (2008) and reviewed in Kuehni (2014).
4. **Public documentation.** The reference implementation's public specification (score-system.md, attribution-exercise design document, companion scientific paper at Colour Research and Application) cites Kuehni (2008) and Kuehni (2014) and states explicitly that the attribution scoring uses the Kuehni mean-observer values rather than the CIECAM02 defaults.

5. Population-specific extension. The claimed method includes the provision to replace the Kuehni (2008) values with population-specific mean-observer values where justified for specific deployment contexts (for example, East Asian, South Asian, or African population samples if and when comparable empirical data become available). The claim is that the scoring pipeline is parameterised to accept empirical mean-observer values as its cardinal anchors, rather than being hard-coded to any specific value set.

Embodiments

The reference implementation uses Kuehni (2008) values as default for deployment in Dutch-EU and comparable Western-European populations. The anchor values are implemented as a configuration constant in the scoring pipeline, permitting substitution with alternative empirical values without recompilation of the instrument.

Alternative embodiments may use: different empirical data sources; population-specific anchor sets selected at session start based on participant self-identified region; or dynamic anchor refinement as larger-scale empirical data accumulate. The narrow claim is specifically the use of empirical mean-observer values (with Kuehni 2008 as the reference set) as cardinal anchors in closed-set attribution classification, implemented as a configurable parameter rather than as a hard-coded constant.

4. CROSS-INNOVATION ARCHITECTURE

The eight innovations disclosed in Sections 3.1 through 3.8 are not independent. They compose into an integrated colour-competency assessment architecture in which each innovation resolves a specific problem that would otherwise compromise the others. This section describes the composition and notes the dependencies.

4.1 Dependency graph

The three-layer competency framework (Section 3.1) is the root of the architecture. It defines what is measured; the remaining innovations address *how* each layer is measured and how the results are reported.

The 3×3 competency matrix (Section 3.2) depends on Section 3.1 directly — the three layers populate the matrix columns. It additionally depends on per-instrument dimension mapping, which in turn depends on the stimulus-generation constraints ensured by Section 3.7.

The differentiated three-track validation strategy (Section 3.3) depends on Section 3.1 — each of the three layers receives its own validation track. It does not depend on the specific instruments used.

The five-segment 18-cap Hue Ordering Diagnostic (Section 3.4) is an instance of a discrimination-layer instrument (Section 3.1 layer a, row hue in Section 3.2 matrix). It depends on the stimulus-generation method of Section 3.7 for its per-cap OKLCH colour construction.

The two-path calibration architecture (Section 3.5) is layer-independent. It applies to any session, regardless of which instruments are administered. It is a precondition for valid measurement in any of the layers and therefore a dependency of Sections 3.1 through 3.4, 3.6, and 3.8.

The cardinal-nearest hue classification (Section 3.6) is the scoring method for the attribution layer on the hue dimension (row hue, column attribution in the Section 3.2 matrix). It depends on Section 3.8 for the specific cardinal-anchor values.

The OKLCH gamut-aware trial generation (Section 3.7) is a stimulus-construction primitive used by Sections 3.4, 3.6, and implicitly by the lightness and chroma ordering diagnostics referenced in Section 3.1.

The Kuehni-anchored closed-set attribution scoring (Section 3.8) provides the specific cardinal-anchor values consumed by Section 3.6.

4.2 Architectural novelty

The individual components have distinct novelty claims, as set out in Sections 3.1 through 3.8. The composition itself has an additional novelty dimension: no known published assessment architecture combines three-layer measurement, 3×3 matrix reporting, differentiated per-layer validation, and two-path calibration in a single deployable digital platform. A patent claim on any single component listed above may be defended by third parties through arguments about prior-art availability of the component in isolation; the composite architecture, as implemented in ColorAptitude and disclosed here, is not available as prior art in the public domain before this publication.

4.3 Relationship to the reference implementation

The ColorAptitude platform, developed by the author and operated through the Nederlandse Kleurenschool and ColorExpertsHub, is the reference implementation of the architecture disclosed in this document. The relationship between the architecture (disclosed as prior art) and the implementation (retained by Xcone and its sublicensees) is as follows:

- The *architecture* — methods, data structures, scoring formulas, reporting formats, calibration protocols, validation strategies — is the subject of this defensive publication and is placed in the public domain as prior art.
- The *implementation* — source code, specific user-interface designs, proprietary datasets (including normative data to be collected), specific server infrastructure, brand marks (ColorAptitude, Xcone, SNKI, Nederlandse Kleurenschool, ColorExpertsHub), and specific integrations with third-party platforms — remains the exclusive property of the author and Xcone, and is not disclosed in this document except to the extent necessary for enabling description of the architecture.

- Third parties may implement the architecture in their own source code, with their own UI, their own datasets, and under their own brand, without infringing on the architecture claims disclosed here. Third parties may not, however, copy the ColorAptitude source code, reuse the Xconea or SNKI brand marks, or access the proprietary datasets.

The distinction between architecture and implementation is central to the defensive-publication strategy. Publication of the architecture prevents patent monopolisation of the architectural ideas. Retention of the implementation preserves commercial value for the author and for Xconea.

5. LEGAL FRAMEWORK AND PUBLICATION RECORD

5.1 Intellectual property ownership and management

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5.2 Nature and purpose of this publication

This document is a defensive publication. Its purpose is the establishment of public prior art for the subject matter disclosed in Sections 3.1 through 3.8. By publishing a dated, attributed, and enabling description of these innovations, the document places the described subject matter into the public domain with respect to novelty considerations in patent examination worldwide. Under 35 U.S.C. § 102(a)(1) and corresponding provisions in the European Patent Convention (Article 54), the European Union Trade Mark Regulation, and comparable instruments in other jurisdictions, publicly available descriptions with an established publication date constitute prior art against subsequent patent claims.

5.3 Scope of disclosure versus scope of retained rights

The **architecture** disclosed in this document — methods, data structures, scoring formulas, reporting formats, calibration protocols, validation strategies — is placed in the public domain as prior art. The **implementation** — source code, specific user-interface designs, proprietary datasets (including normative data to be collected), specific server infrastructure, brand marks, and specific integrations with third-party platforms — remains the exclusive property of the author and Xconea. Third parties are free to implement the disclosed architecture in their own systems, without infringing the architecture claims set out here. Third

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5.6 Version and integrity

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5.8 Warranties and contact

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For questions regarding the subject matter disclosed, or for licensing enquiries relating to the ColorAptitude reference implementation, correspondence may be directed to m.kotterink@kleurinstituut.nl.

REFERENCES

- Fairchild, M. D. (2018). Unique hues and principal hues. *Color Research and Application*, 43(5), 760–765.
- Farnsworth, D. (1957). *The Farnsworth-Munsell 100-Hue Test for the examination of color discrimination* (rev. ed.). Munsell Color Company.
- Hardy, J. L., Frederick, C. M., Kay, P., & Werner, J. S. (2005). Color naming, lens aging, and grue: What the optics of the aging eye can teach us about color language. *Psychological Science*, 16(4), 321–327.
- Hirschler, R., Gay, J., & Ferreira de Oliveira, D. (2002). Field trials of three tests for colour vision and colour aptitude. In R. Chung & A. Rodrigues (Eds.), *Proceedings of SPIE, Vol. 4421: 9th Congress of the International Colour Association* (pp. 323–326). SPIE.
- Hunt, R. W. G. (2004). *The reproduction of colour* (6th ed.). Wiley / IS&T.
- International Organization for Standardization. (2009). *ISO 3664:2009: Graphic technology and photography — Viewing conditions*. ISO.

Kotterink, M. (2026, in preparation). *A digital platform for professional colour competency: Assessment, training and certification*. Manuscript prepared for submission to Colour Research and Application.

Kuehni, R. G. (2008). Experimental object color unique hue data for the mean observer. *Color Research and Application*, 33(6), 505–506.

Kuehni, R. G. (2014). Unique hues and their stimuli — state of the art. *Color Research and Application*, 39(3), 279–287.

Logvinenko, A. D. (2012). A theory of unique hues and colour categories in the human colour vision. *Color Research and Application*, 37(2), 109–116.

Ottosson, B. (2020). *A perceptual color space for image processing*. <https://bottosson.github.io/posts/oklab/>

Siuda-Krzywicka, K., Witzel, C., Chabani, E., Taga, M., Coste, C., Cools, N., Ferrieux, S., Cohen, L., & Bartolomeo, P. (2019). Color categorization independent of color naming. *Cell Reports*, 28(10), 2471–2479.

Skelton, A. E., Catchpole, G., Abbott, J. T., Bosten, J. M., & Franklin, A. (2017). Biological origins of color categorization. *Proceedings of the National Academy of Sciences*, 114(21), 5545–5550.

American Society for Testing and Materials. (2016). *ASTM E1499-16: Standard Guide for the Selection, Evaluation, and Training of Observers*. ASTM International.

American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. American Educational Research Association.

Wyszecki, G., & Stiles, W. S. (1982). *Color science: Concepts and methods, quantitative data and formulae* (2nd ed.). Wiley.

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